

Reciprocal $E \rightarrow I$ coupling captures phase in large PING circuits

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Abstract

Reciprocal long-range excitation can capture the phase of two independently driven PING circuits containing 800 excitatory and 200 inhibitory neurons each in a narrow strength–delay region. Excitatory neurons in each circuit project to the other circuit’s inhibitory population, which then inhibits its local excitatory population. The uncoupled pair oscillates at 41.504 Hz and 37.842 Hz, a 9.23% separation, with phase-locking value 0.248 and 3.662 Hz residual frequency difference. A registered coarse sweep found no condition passing every joint gate. A declared exploratory refinement then found a contiguous neighborhood with zero frequency difference and high phase locking. Its strongest condition, $E \rightarrow I$ weight 0.25 and delay 10 ms, reaches PLV 0.899, retains half-window PLVs 0.919 and 0.88, and limits half-window phase drift to 0.084 rad. A stronger test continues a mature saved state once with static weight 0.22 coupling. It reaches the sustained-250-ms capture criterion after 922.5 ms. The coupled phase is initially 0.38 rad from its eventual lag, but its final-quarter median error falls to 0.205 rad. During the final 250 ms, 95% of the smoothed instantaneous frequency differences are below 1.58 Hz. This trajectory therefore shows acquisition across the observation window rather than beginning already synchronized. Strong coupling instead suppresses one circuit. This one-seed result is a reduced mechanism demonstration, not an estimate of biological generality.

Methods

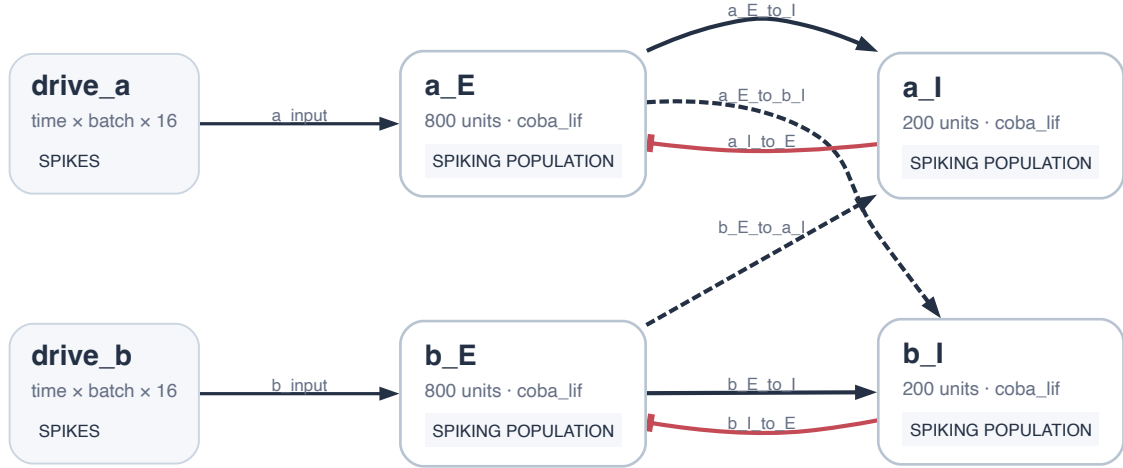


Figure 1: Experimental topology. Independent spike inputs drive circuits A and B, each containing 800 excitatory and 200 inhibitory neurons. Dashed AMPA projections run from each E population to the other circuit’s I population; local I→E projections close each PING loop. There is no shared input or direct cross-circuit inhibition.

1. **Construct the oscillator pair.** The hypothesis was that moderate reciprocal excitatory-to-inhibitory (E→I) AMPA coupling would entrain two active, independently driven PING circuits by recruiting inhibition in the other circuit. Zero coupling would permit drift, whereas excessive coupling could suppress activity. Each circuit contains 800 excitatory and 200 inhibitory conductance neurons. Independent Poisson generators drive 16 channels per circuit. Their seeds are archived separately, so common-input synchrony is impossible. Every condition uses seed 78, a 0.1 ms timestep, and a 1800 ms duration. Analysis excludes the first 400 ms.

The superseded run connected each inhibitory population directly to the other circuit’s excitatory population. That shortcut did not represent the intended long-range anatomy, so its results were discarded. Before this replacement sweep, the cross-circuit pathway was amended to E→I while the uncoupled calibration, input realization, strength and delay grids, initial conditions, analysis, and acceptance thresholds remained fixed.

A subsequent scale amendment replaced the preliminary 40 E / 10 I circuits with the 800 E / 200 I circuits reported here. Calibration and the complete coupling sweep were rerun at the larger scale; none of the preliminary small-circuit measurements enters this report.

2. **Calibrate detuned gamma activity.** The bounded calibration changed input rate and input weight only; all within-circuit excitatory–inhibitory settings remained fixed. A candidate had to retain finite, non-silent, non-saturated activity, a prominent 30–80 Hz rate peak in each circuit, and 5–15% detuning. The deterministic score selected the valid candidate nearest 10% detuning, with peak prominence and grid order breaking ties. Cross-coupling was not evaluated during calibration.
3. **Apply the registered locking gate.** Locking required both circuits to remain active, frequency difference no more than 50% of baseline, phase-locking-value (PLV) gain at least

0.15, mean-band coherence gain at least 0.1, PLV at least 0.55 in each half-window, and phase-offset drift no more than 0.6 rad. PLV ranges from zero for dispersed phase differences to one for a fixed phase relationship. A passing condition also needed an orthogonally adjacent locked condition. Silent conditions were never eligible.

4. **Refine the observed boundary.** After the registered sweep failed, an explicitly exploratory grid sampled six weights from 0.10 through 0.35 and six delays from 8 through 15 ms. Conditions were ranked, without altering the registered result, by normalized frequency difference, full-window PLV, minimum half-window PLV, and half-window phase drift. The best condition was selected mechanically before the comparison figure was drawn.
5. **Continue a mature phase-separated state.** Cross-circuit projections were present at weight zero during burn-in, making the zero-weight and coupled graphs structurally state-compatible. Independent Poisson inputs evolved both circuits to timestep 11662 (1166.2 ms). The zero-coupling run was repeated exactly to that timestep and complete membrane, refractory, conductance, recurrent-delay, and delayed-input state was exported. The same saved state and future input realization then initialized one static reciprocal E→I coupling continuation. No uncoupled continuation was simulated. After the broader weight–delay refinement, a bounded 2 s search compared mature checkpoints and weights 0.20–0.30 on one fixed archived input stream at delay 10 ms. The declared gate required early PLV at most 0.90, displaced early phase, delayed acquisition, late PLV at least 0.92, 95% of final-250-ms circular phase errors below 0.65 rad, and 95% of final-250-ms instantaneous frequency differences below 3 Hz; the passing case with the smallest late phase error was selected. Weights were not part of runtime state. A 250 ms pre-checkpoint recording context was used only to prevent band-pass/Hilbert edge bias; no spikes were shifted or replayed into the model.

Calibration produced active detuned gamma

Diagnostic	Circuit A	Circuit B
Dominant frequency	41.504 Hz	37.842 Hz
Peak prominence	206.5	207.8
Excitatory rate	38.053 Hz	35.622 Hz
Inhibitory rate	116.429 Hz	110.704 Hz
E active-cell fraction	1	1
I active-cell fraction	1	1

Candidate 9 used input weight 0.55 and independent per-channel drive rates 180 Hz and 150 Hz. All cells participated. The first analysis implementation took the maximum coherence anywhere in the gamma band. The baseline saturated near one, making the registered gain impossible. Before any coupling condition ran, the coherence statistic was frozen as the arithmetic mean of magnitude-squared coherence bins from 30 through 80 Hz. The corrected uncoupled baseline is 0.578.

Mature-state continuation reveals phase capture

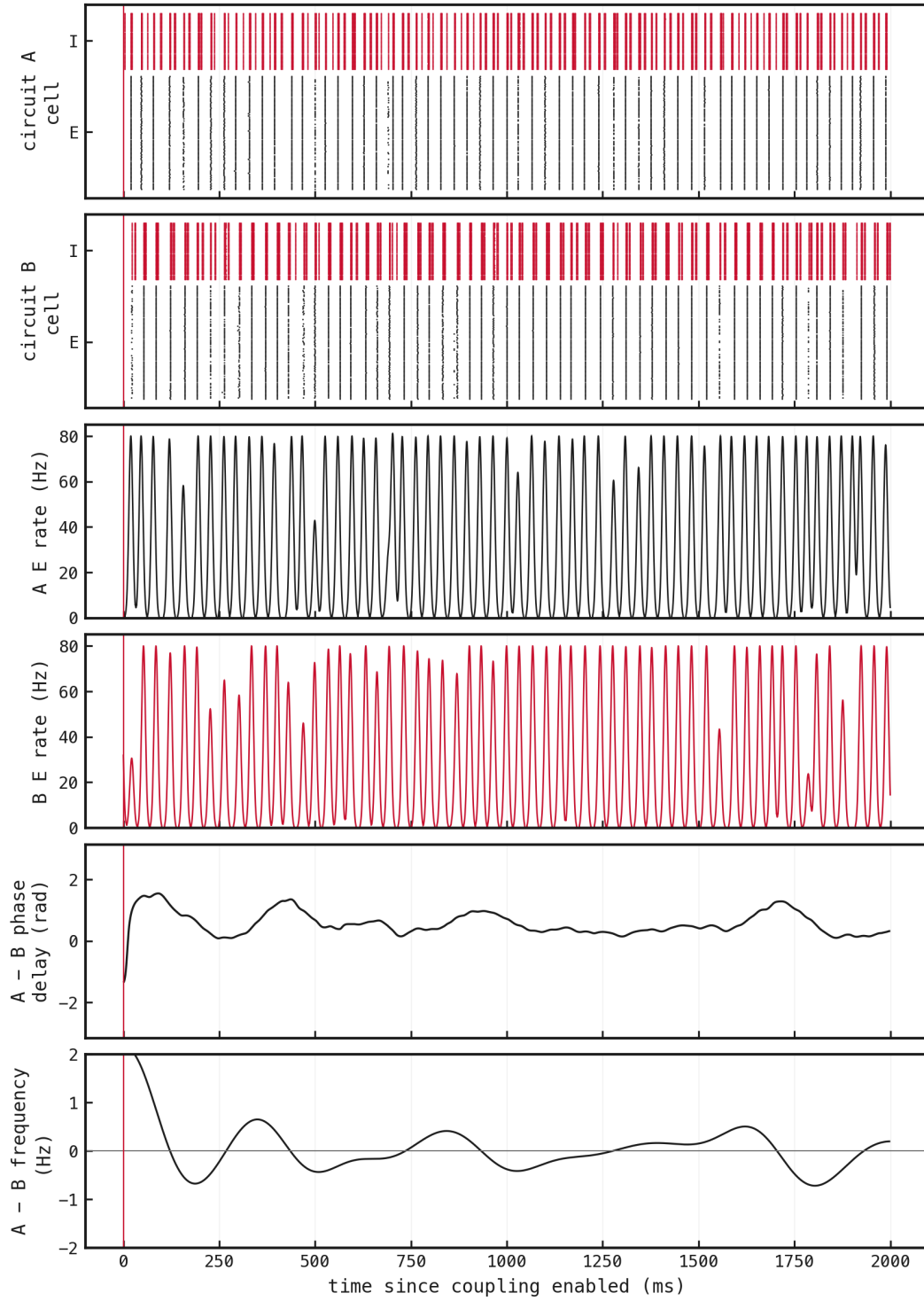


Figure 2: One continuous coupled trajectory from an already mature checkpoint. Reciprocal $E \rightarrow I$ weight 0.22 and delay 10 ms are enabled at 0 ms; nothing else is reset. The first two panels show fixed samples of 100 E and 50 I neurons from circuits A and B. The next two show their full-population E rates on a common scale without overlap. The final two panels show 150 ms circularly smoothed A-minus-B phase delay and the corresponding instantaneous frequency difference, additionally smoothed over 75 ms. Phase wanders over the early and middle window while frequency difference changes sign. After 922.5 ms, phase remains bounded around a preferred lag and slow frequency difference remains centred near zero.

Condition	Frequency difference	PLV	30–80 Hz coherence	Phase (rad)	Lag (ms)
Uncoupled	3.662 Hz	0.248	0.578	−1.341	5.5 ms
0.2, short	2.441 Hz	0.653	0.598	−2.874	−18.2 ms
0.2, half period	0 Hz	0.698	0.527	0.42	31.7 ms
0.5, intermediate	4.883 Hz	0.787	0.638	−0.005	−29.7 ms
0.25, 10 ms exploratory	0 Hz	0.899	0.589	0.181	33.3 ms

The coarse registered sweep remains a negative result: none of its 15 coupled cells passes every joint gate. The subsequent 36-cell exploratory refinement samples weights 0.10–0.35 and delays 8–15 ms. Seven cells have zero frequency difference and PLV above 0.8, including adjacent cells at weights 0.25 and 0.30 and delays 8–11.5 ms. The selected condition has equal 30.518 Hz spectral peaks, PLV 0.899, and only 0.084 rad phase-offset change between analysis halves. Its mean broad-band coherence does not meet the original +0.10 gain gate. That gate averages the whole 30–80 Hz band and is poorly matched to a pair that collapses onto a narrow 30.5 Hz rhythm; it is retained as provenance, but does not negate the direct frequency and phase evidence for synchronization.

Complete-state continuation changes the mechanistic interpretation. The checkpoint begins with two healthy gamma oscillators rather than two freshly reset populations, and it contains no imposed spike-stream offset. Under weight 0.22 coupling, median phase error across the four successive analysis quarters is 0.588, 0.462, 0.234, and 0.205 rad. The smoothed late-phase span is 0.536 rad around an eventual lag of 0.207 rad, while the final-window 95th percentile absolute frequency difference is 1.58 Hz. Coupling therefore changes the phase dynamics over repeated cycles rather than merely selecting an already locked initial condition. The longer window also prevents an overclaim: post-capture PLV is 0.883, and 95% of post-capture phase errors lie below 1.164 rad. The relationship is bounded and frequency-synchronized on the displayed slow timescale, but it is not an absorbing fixed-phase state.

Strong coupling suppresses one circuit

At strength 2 and short delay, circuit B has 0 Hz excitatory and 296.429 Hz inhibitory activity, while circuit A remains at 38.053 Hz and 116.429 Hz. The condition is explicitly inactive and spectrally invalid. Other strong short- and intermediate-delay conditions show the same one-sided suppression boundary. Half-period delay avoids complete silence at the largest registered strengths, but those cells fail the full locking gates.

Conclusion and limitations

The registered coarse-grid success criterion fails, but the explicitly exploratory refinement reveals the missed phase-locking region. Reciprocal E→I excitation can reshape relative phase over many cycles and produce a bounded phase relationship with near-zero slow frequency difference. The 2 s extension retains noisy phase corrections rather than a permanent fixed lag. Excessive coupling recruits enough inhibition to suppress circuit B rather than

synchronize it. This is the intended reduced mechanism: long-range excitation can synchronize PING circuits by recruiting the other circuit's local inhibition.

The evidence has one exploratory seed, one calibrated oscillator pair, dense connectivity, and one set of intrinsic PING parameters. It does not estimate robustness across seeds, circuit sizes, detuning ranges, input statistics, or biological preparations.